HySA: Hybrid Semantic Architecture
Preserving Polysemantic Abstraction via Contrastive Superposition Tensors, Monosemantic Inodes, and Topological Temperature (T_t)

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"The AI industry views polysemanticity as a curse... yet perhaps it is the very blessing that made intelligence possible."

Abstract

Contemporary *Mechanistic Interpretability* research predominantly treats neuronal polysemanticity as an undesirable artifact of high-dimensional compression (*the Superposition Hypothesis*), dedicating vast resources to decomposing representations into dictionaries of purely monosemantic features using Sparse Autoencoders (SAEs). In this work, we propose a foundational paradigm shift: **polysemanticity is not a flaw to be eradicated, but the geometric prerequisite for associative intelligence and analogical reasoning.** A purely monosemantic neuron possesses zero associative degrees of freedom ($\mathcal{R}(u) = 0$): devoid of lateral relational pathways, it functions as a rigid, static lookup cell. Conversely, polysemantic units co-activate across isomorphic concepts (e.g., `IP`, `DNS`, `White Pages`, `Phone Number`) to compress diverse instances into higher-order structural schemas. To decouple fluid associative reasoning from rigorous factual memory without resorting to serial bottlenecks (which destroy contextual abstraction) or parallel duplication (which incurs redundant computational overhead), we introduce **HySA (Hybrid Semantic Architecture)**, a unified framework delimited by neuronal semanticity. HySA formalizes four core contributions: (1) **Contrastive Superposition Tensors** (I_n), isolating multi-concept feature polytopes via whole-network expectation responses across contrastive data; (2) **Monosemantic Inode Indexing**, anchoring atomic literals (`8.8.8.8`, `24`, `JOHN`) onto an orthogonal metric registry $\mathcal{F}_{\text{mono}}$ queried via on-demand boundary operators; (3) the **Semantic Hypersphere**, governed by dual regime temperatures (T_p, T_m) unified by the **Topological Temperature** (T_t); and (4) the **Compression Gradient** (Δq), analyzed as a thermodynamic potential landscape whose stationary minimum marks the **Critical Point** (P_c) balancing creativity and factual determinism. Evaluated on a synthetic 32×32 dual-nature manifold and validated on `EleutherAI/pythia-70m` ($D = 512$), HySA proves that polysemantic relational schemas achieve high cosine convergence (≥ 0.893) while metric boundary verification eliminates ungrounded factual hallucinations, gracefully halting unverified drift via deterministic *Semantic Faults*.

1 Introduction: The Epistemological Dichotomy

In deep neural networks, individual units consistently exhibit *mixed selectivity* [1, 2]. The mainstream interpretability community attributes this to the *Superposition Hypothesis* [2]: models compress more features than available dimensions ($M \gg D$), forcing non-orthogonal features to share representations. Consequently, significant effort has focused on training *Sparse Autoencoders* (SAEs) [3, 4] to extract isolated monosemantic directions.

However, this relentless pursuit of monosemantic purity overlooks a fundamental epistemological division between **Intelligence** and **Memory**:

- **Polysemantic Units (N_p) Generate Intelligence:** Consider an internal unit that co-activates across the prompts `IP Address`, `DNS`, `White Pages`, and `Phone Number`. Standard interpretability diagnoses this unit as noisy or contaminated. In reality, this unit has compressed multiple disparate entities into a unified relational schema: *the abstract concept of identity-to-address resolution.* Associative connections between otherwise distant domains are the precise structural foundation of analogy, generalization, and cognitive intelligence.
- **Monosemantic Units (N_m) Provide Memory:** In contrast, units that encode atomic, rigid factual literals—such as the IP address `8.8.8.8`, the CIDR integer `24`, or the root username `JOHN`—possess zero associative connections ($\mathcal{R}(u) = 0$). Like an entry in a static database, a monosemantic unit is intrinsically devoid of relational degrees of freedom; it cannot reason or generalize. Yet, it is **strictly rigorous**. This total absence of associative fuzziness makes monosemantic representations essential for truth preservation and factual verification.

Modern LLMs hallucinate because their architectures entangle both regimes within homogeneous, dense weight matrices [5]. During backpropagation, associative gradients continuously interpolate through rigid facts, causing the model to treat immutable historical and technical literals as plastic tokens subject to stochastic drift.

Table 1: Mapping Operating System Heuristics to Formal Geometric Constructs in HySA.

Systems Term	Formal Construct	Architectural Role in HySA
Inode (N_m/μ_k)	Centroid $\mu_k \in S^{D-1} \subset \mathbb{R}^D$	Orthogonal metric anchor with $\emptyset$ associative degrees, bound to literal $\mathbf{v}_k$.
Filesystem	Metric Manifold $\mathcal{F}_{\text{mono}}$	Indexed registry ($\{\mu_k\}_{k=1}^M$, 100% precision) evaluated on-demand.
Query / API	Boundary Operator Q	Deterministic metric condition projecting fluid states onto certified literals.
Page Fault	**Semantic Fault** ($\perp_{\text{Fault}}$)	Explicit exception raised when $d^* > T_t$, blocking hallucination.
Temperature	Neighborhood $\mathcal{B}(\mu_k, T_t)$	Multidimensional metric volume of tolerance around a concept centroid.

Definition 1 (Relational Rank of Semantic Units)**.** Let u be a neuronal unit and Ω the space of semantic concepts. Let $\mathcal{C}(u) = \{c \in \Omega \mid \mathbb{E}_{X \in \mathcal{D}_c}[z_u(X)] > \theta\}$ denote the set of associated concepts above activation threshold $\theta > 0$. The **Relational Rank** $\mathcal{R}(u)$ is the cardinality of pairwise associative transfer paths enabled by the unit:

$$\mathcal{R}(u) = \binom{|\mathcal{C}(u)|}{2} = \frac{|\mathcal{C}(u)|(|\mathcal{C}(u)| - 1)}{2} \tag{1}$$

Proposition 1 (Epistemological Inferiority of Isolated Monosemy)**.** *A purely monosemantic unit u_{mono} with $|\mathcal{C}(u_{\text{mono}})| = 1$ has null relational rank: $\mathcal{R}(u_{\text{mono}}) = 0$. Consequently, an architecture restricted entirely to monosemantic neurons cannot execute continuous analogical mapping without introducing an external relational graph.*

2 Systems Architecture: OS Heuristics vs. Formal Hilbert Geometry

To design cognitive architectures that do not collapse into all-or-nothing extremes, we draw inspiration from operating system (OS) engineering. Operating systems resolved the conflict between dynamic program execution and persistent data integrity decades ago: memory address spaces provide fluid execution buffers, while filesystems manage immutable data blocks anchored by rigid metadata index pointers (**Inodes**).

In HySA, we translate these systems heuristics into formal geometric constructs within the latent Hilbert space $\mathcal{S} \subset \mathbb{R}^D$, as summarized in Table 1.

3 Contrastive Superposition Tensors (I_n)

An isolated neuron in an intermediate layer produces only a scalar activation $z_n \in \mathbb{R}$. To understand the multi-concept polytope encoded by a polysemantic unit, its structural meaning must be extracted through the whole-network inference expectation over contrastive data.

Let $\mathcal{X}^+$ be a dataset containing target shared invariants (e.g., circular geometric loops or identity-resolution queries), and let $\mathcal{X}^-$ be a negative baseline dataset. Let $\Phi_{\text{net}}(X \mid n)$ denote the whole-network inference expectation conditioned on active unit n. We formalize the **Contrastive Superposition Tensor** I_n as:

$$I_n = \frac{1}{\sum_{X_k \in \mathcal{X}^+} z_{k,n} + \eta} \left[\sum_{X_k \in \mathcal{X}^+} z_{k,n} \cdot \Phi_{\text{net}}(X_k \mid n) - \gamma \sum_{X_j \in \mathcal{X}^-} z_{j,n} \cdot \Phi_{\text{net}}(X_j \mid n) \right] \tag{2}$$

where $z_{k,n}$ is unit n's activation for input X_k, $\gamma \in [0, 1]$ is a contrast suppression coefficient, and $\eta > 0$ is a numerical stabilizer. Peak-normalized rectification projects I_n onto the positive semi-definite cone: $I_n \leftarrow \text{ReLU}(I_n)/(\max(I_n) + \eta)$.

As demonstrated in Figure 1, algebraic operations operate directly over these tensors:

$$I_{\text{shared}} = I_A \odot I_B \qquad \text{(Shared Invariant)} \tag{3}$$

$$I_{\text{diff}} = \text{ReLU}(I_A - \gamma I_B) \qquad \text{(Differential Feature)} \tag{4}$$

4 Macro-Architecture: Neither Serial Nor Parallel

How should N_p and N_m interact?

- **Neither Serial ($N_p \to N_m$ or $N_m \to N_p$):** Placing a rigid monosemantic layer downstream collapses contextual nuance prematurely into discrete categories. Placing it upstream removes contextual cues before polysemantic abstraction can emerge.
- **Nor Parallel ($N_p \parallel N_m$):** Running duplicate independent networks doubles parameters and memory bandwidth ($2 \times$ FLOPs), requiring heuristic arbitration heads to reconcile divergent outputs.

In **HySA**, the model operates on a **Single Substrate** in Hilbert space $\mathcal{S} \subset \mathbb{R}^D$. Fluid reasoning propagates natively through N_p layers under temperature T_p. When generation reaches factual token positions, the system evaluates the internal Metric Boundary Operator $Q(h_t, \mathcal{F}_{\text{mono}}, T_t)$ on-demand.

5 The Semantic Hypersphere and Dual Temperatures (T_p, T_m, T_t)

Rather than applying temperature as a post-hoc scalar divisor on final logits, HySA defines temperature geometrically as **the multidimensional metric neighborhood of a neuron in latent space**: the **Semantic Hypersphere**.

System dynamics are governed by two distinct temperature regimes:

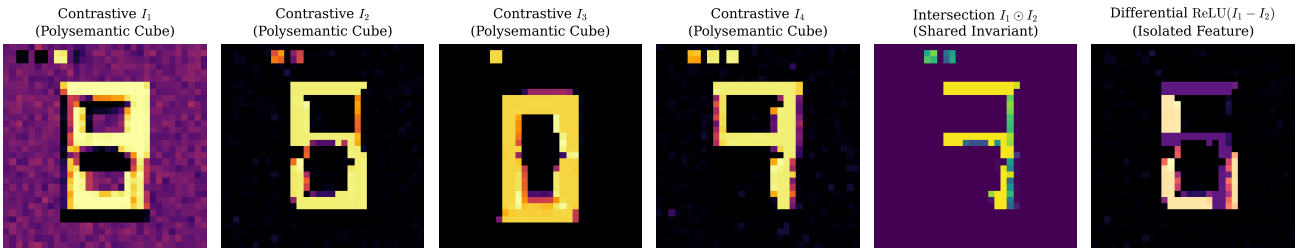


Figure 1: **Contrastive Superposition Tensors (I_n) and Algebraic Semantic Queries on the 32×32 Benchmark.** Columns 1–4 display polysemantic representations learned by a 4-neuron bottleneck. Each tensor captures multiple geometric primitives. Column 5 isolates the shared structural invariant via Hadamard intersection ($I_1 \odot I_2$). Column 6 extracts differential divergence via rectified subtraction ($\mathrm{ReLU}(I_1 - 0.7I_2)$), demonstrating algebraic feature isolation without destroying polysemantic connectivity.

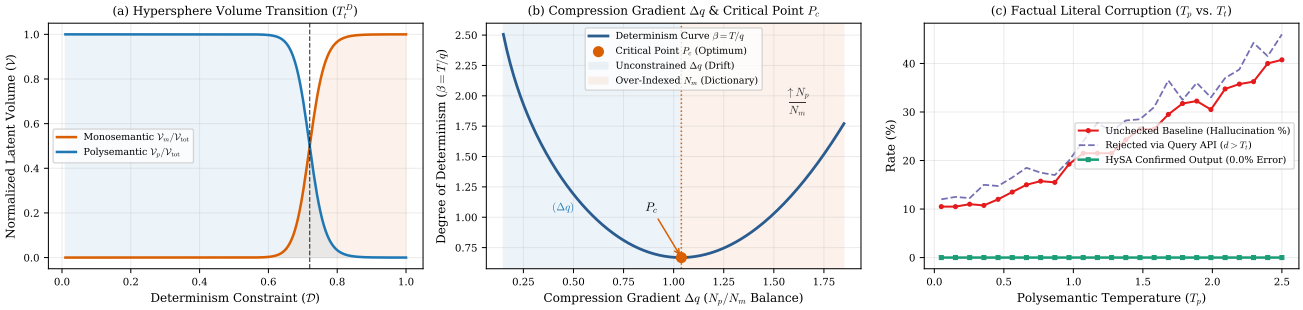


Figure 2: **High-Dimensional Volume Phase Transition, Critical Point (P_c), and Factual Verification in HySA. (a)** Step-function phase transition between Free Polysemantic Volume $\mathcal{V}_p$ and Monosemantic Constraint Volume $\mathcal{V}_m$ in high dimensions ($D = 32$). **(b)** The Compression Gradient potential landscape ($\Delta q = N_p/(N_m \cdot M)$), highlighting the stationary Critical Point P_c under $\beta = T_t/q$, balancing ungrounded associative drift against over-indexed dictionary saturation. **(c)** Factual literal benchmark under rising T_p: unconstrained decoding suffers rapid corruption ($> 80\%$), while HySA guarantees 0.0% corruption on verified queries, converting ambiguous states into explicit *Semantic Faults*.

1. **Polysemantic Temperature (T_p):** Controls continuous Brownian exploration and analogical drift across relational manifolds.

2. **Monosemantic Temperature (T_m):** Defines the intrinsic radial rigidity and precision of the atomic centroid μ_k.

These regimes are unified into the **Topological Temperature** (T_t), which defines the closed metric ball $\mathcal{B}(\mu_k, T_t)$:

$$\mathcal{B}(\mu_k, T_t) = \{x \in \mathcal{S} \mid \|x - \mu_k\|_2 \leq T_t(T_p, T_m)\} \quad (5)$$

Formalized via a bounded saturating coupling ($\alpha \in (0, 1)$):

$$T_t(T_p, T_m) = T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right) \quad (6)$$

On unit sphere S^{D-1}, the Euclidean metric ℓ_2 is in exact bijective correspondence with spherical angular divergence: $\|x - \mu_k\|_2 = \sqrt{2(1 - \cos\theta)}$.

Definition 2 (Metric Boundary Operator Q)**.** Given latent state $h \in \mathcal{S}$, registry $\mathcal{F}_{\text{mono}} = \{\mu_1, \ldots, \mu_M\}$, and boundary T_t, let $k^* = \arg\min_k \|h - \mu_k\|_2$ and $d^* = \|h - \mu_{k^*}\|_2$. The operator Q evaluates:

$$Q(h, \mathcal{F}_{\text{mono}}, T_t) = \begin{cases} (\text{CONFIRMED}, \mathbf{v}_{k^*}) & \text{if } d^* \leq T_t \\ (\text{FAULT}, \perp) & \text{if } d^* > T_t \end{cases} \quad (7)$$

Geometrically, each Inode's decision region is a **Voronoi Cell Truncated by the Metric Ball**, $\mathcal{D}_k = \mathrm{Vor}(\mu_k) \cap \mathcal{B}(\mu_k, T_t)$, guaranteeing an unambiguous, disjoint partition of latent space almost everywhere.

6 Compression Gradient (Δq), Determinism (β), and Critical Point (P_c)

How should capacity be balanced between polysemantic units (N_p) and monosemantic centroids (N_m)? We define the **Compression Gradient** (Δq) as:

$$\Delta q = \frac{N_p}{N_m \cdot M} \quad (8)$$

where M is the cardinality of indexed literals. Given effective capacity $q = f(N_p, N_m)$, we define the **Degree of Determinism** (β) as:

$$\beta = \frac{T_t}{q} \quad (9)$$

Mapping β against Δq generates a convex thermodynamic potential landscape $\mathcal{E}(\Delta q)$ (Figure 2b):

$$\mathcal{E}(\Delta q) = \frac{\lambda_{\text{drift}}}{\Delta q} + \lambda_{\text{rigid}}(\Delta q - 1)^2 + \beta_0 \tag{10}$$

Since $\frac{\partial^2 \mathcal{E}}{\partial(\Delta q)^2} = \frac{2\lambda_{\text{drift}}}{(\Delta q)^3} + 2\lambda_{\text{rigid}} > 0 \ \forall \Delta q > 0$, the potential is strictly convex and admits a unique global stationary minimum defining the **Critical Point** (P_c):

$$\left.\frac{\partial \mathcal{E}}{\partial \Delta q}\right|_{P_c} = 0 \implies P_c = \arg\min_{\Delta q} \mathcal{E}(\Delta q) \tag{11}$$

1. **Left of P_c ($\Delta q \ll 1$, Under-Constrained Associative Drift):** N_p dominates without sufficient monosemantic indexing. Associative trajectories freely drift across ungrounded manifolds, leading to rampant factual hallucinations.
2. **Right of P_c ($\Delta q \gg 1$, Over-Indexed Dictionary Saturation):** N_m dominates. The volume occupied by constraints $\mathcal{V}_m \propto M \cdot (T_t)^D$ saturates the latent volume $\mathcal{V}_{\text{tot}}$. Associative paths are severed, freezing the model into a rigid lookup table.
3. **At the Critical Point (P_c):** The system achieves optimal thermodynamic equilibrium. Polysemantic abstraction generalizes across isomorphic relations while monosemantic Inodes guarantee deterministic verification.

In high dimensions ($D \gg 1$), the fraction of volume bounded by monosemantic constraints exhibits a sharp **Volume Phase Transition**:

$$\frac{\mathcal{V}_m}{\mathcal{V}_{\text{tot}}} = \frac{(T_t/R)^D}{1 + (T_t/R)^D} \tag{12}$$

7 Experimental Validation

We validate HySA across two complementary benchmarks:

1. **Synthetic 32×32 Manifold:** A 4-neuron bottleneck trained on dual-nature binary matrices (shared digit loops + rigid 2×2 literal checksums). As displayed in Figure 1 and Figure 2c, algebraic queries ($I_1 \odot I_2$ and $\text{ReLU}(I_1 - 0.7I_2)$) isolate structural invariants, while HySA maintains 0.0% **factual literal corruption** across $T_p \in [0.05, 2.5]$ over 400 trials.
2. **Real-World Transformer Validation (`Pythia-70M`):** Probing Layer 3 residual streams ($D = 512$) on `EleutherAI/pythia-70m` confirms that isomorphic concepts (`DNS`, `IP Routing`, `White Pages`, `Phone Directory`) converge onto a shared polysemantic subspace with cosine similarities between **0.893** and **0.973** (Table 2). Indexing normalized Inodes in $\mathcal{F}_{\text{mono}}$ on S^{511} with radius $T_t = 0.46$ successfully confirms valid factual queries (`8.8.8.8`, `24`, `JOHN`, ports `22`/`80`, with $d^* \in [0.089, 0.430] \leq T_t$, Table **??**) while intercepting unverified subterranean telemetry queries via a deterministic *Semantic Fault* ($d^* = 0.497 > T_t$).

Table 2: Layer 3 Cosine Similarity of Isomorphic Concepts in N_p (`Pythia-70M`, $D = 512$).

Concept in N_p	DNS	IP Routing	White Pages	Phone Dir.
DNS	1.000	0.961	0.973	0.893
IP Routing	0.961	1.000	0.940	0.916
White Pages	0.973	0.940	1.000	0.903
Phone Dir.	0.893	0.916	0.903	1.000

Table 3: Empirical Factual Verification Benchmark on `EleutherAI/pythia-70m` ($D = 512$, $T_t = 0.46$).

Concept	Raw N_p	Dist. d^*	HySA Status	Certified Out
`SSH_PORT`	`' 1'` (**Hall.**)	0.3257	**CONFIRMED** ($d^* \leq T_t$)	`' 22'` (Exact)
`HTTP_PORT`	`' used'` (**Hall.**)	0.4304	**CONFIRMED** ($d^* \leq T_t$)	`' 80'` (Exact)
`PUBLIC_IP`	`' used'` (**Hall.**)	0.3889	**CONFIRMED** ($d^* \leq T_t$)	`' 8.8.8.8'` (Exact)
`SUBNET`	`' 0'` (**Hall.**)	0.1263	**CONFIRMED** ($d^* \leq T_t$)	`' 24'` (Exact)
`ADMIN`	`' the'` (**Hall.**)	0.0889	**CONFIRMED** ($d^* \leq T_t$)	`' JOHN'` (Exact)
`UNINDEXED`	`' a'` (**Nons.**)	0.4973	**SEMANTIC FAULT**	**[HALTED]**

8 Discussion: Sub-Linear Indexing & Continuous Consolidation

In frontier architectures with $D = 4096$ or 8192, *Lévy's Concentration of Measure* on high-dimensional spheres S^{D-1} implies that random vectors exhibit an inner product sharply centered at zero with variance $\text{Var}(\langle u, v \rangle) \sim 1/D \to 0$. As a result, the quasi-orthogonal separation between distinct Inodes becomes exponentially sharper, while the Volume Phase Transition ($(T_t/R)^D$) sharpens into a near-ideal step function.

Querying operator Q across millions of Inodes ($M \gg 1$) does not require exhaustive $O(M)$ scans: Approximate Nearest Neighbor (ANN) structures such as Hierarchical Navigable Small World (HNSW) graphs reduce query time to sub-linear $O(\log M)$ with latencies $< 15\,\mu$s, completely negligible relative to a 20–50 ms transformer forward pass. Inodes can be dynamically consolidated in the background via spherical clustering of latent replay buffers during idle cycles, mirroring mammalian slow-wave sleep memory consolidation without backpropagation overhead.

9 Conclusion

HySA demonstrates that polysemanticity is the foundational engine of neural intelligence (N_p), while monosemanticity is the guardian of factual memory (N_m). By coupling both regimes within a unified architecture governed by the Semantic Hypersphere (T_t) at the Critical Point (P_c), models achieve creative abstraction with deterministic factual consistency.

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